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Section 01

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Table of Contents

Chapter 1: Introduction	4
1.1 Overview	4
1.2 Problem Statement	4
1.3 Project Objectives	4
1.4 Project Scope.....	5
1.4.1 In-Scope	5
1.4.2 Out of Scope.....	6
1.5 Project Importance	6
Chapter 2: Literature Review	7
2.1 Introduction	7
2.2 Enterprise Architecture (EA).....	7
2.2.1 Definition and Concepts of EA	7
2.2.2 Review of Existing Studies Related to University Enterprise Architecture	8
2.3 Related Sub-System	12
2.3.1 Overview of the chosen sub-systems	12
2.4 Technology Used.....	14
2.4.1 Unified Data Management and Analytics	14
2.4.2 Application Development and Integration	14
2.4.3 Extension of Enterprise Systems	14
2.4.4 Alignment with EA Frameworks.....	14
2.5 Summary	15
Chapter 3 Methodology.....	16
3.1 Introduction	16
3.2 Chosen Methodology	16
3.3 Phases of Methodology	16
3.3.1 Planning Phase	16
3.3.2 Analysis Phase.....	16
3.3.3 Design Phase	17

3.3.4 Implementation Phase	17
3.3.5 Testing Phase.....	17
3.4 Project Planning Schedule.....	17
3.5 Summary	17
Chapter 4: Analysis and Design	19
4.1 Introduction	19
4.2 Organization Structure	19
4.3 Current System Analysis	20
4.3.1 Computerized Process and Infrastructure.....	20
4.3.2 Manual Processes and Governance	21
4.3.3 Identified System Issues	21
4.4 Comparison between Existing and Proposed System	21
4.4.1 Current Digital Infrastructure	21
4.4.2 Operational Challenges	21
4.4.3 Proposed System Enhancement.....	22
4.4.4 Expected Outcomes	22
4.5 System Requirements Gathering Techniques.....	23
4.5.1 Daily Operations & Current Process	23
4.5.2 Enterprise Architecture & Integration	23
4.5.3 Infrastructure & Performance.....	23
4.5.4 User Experience & Future Needs	23
4.8. System Design.....	23
4.8.1 Enterprise architecture EA-AS IS	24
4.8.2 Explain each component in system architecture EA-AS IS	24
4.8.3 Enterprise architecture EA-TO BE.....	29
4.8.4 Explain each component in system architecture EA- TO BE	29
4.8.3 System Architecture	35
4.8.4 System Architecture Component Explanation	36
4.8.5 Project Design	38
4.8.5.1 Use Case Diagram - Payment & Billing Tracker	38

4.8.5.2 Use Case Description - Payment & Billing Tracker.....	38
4.8.5.3 Activity Diagram - Payment & Billing Tracker	40
4.8.5.4 Sequence Diagram - Payment & Billing Tracker.....	41
4.8.6 Database Design - Payment & Billing Tracker	42
4.8.6.1 Class Diagram - Payment & Billing Tracker	42
4.8.6.2 Entity Relationship diagram - Payment & Billing Tracker	42
4.8.7 Interface Design - Payment & Billing Tracker.....	43
4.9 Summary	43

Chapter 1: Introduction

1.1 Overview

University Tun Hussein Onn Malaysia (UTHM) is a public university that provides various academic, administrative and support services to students. These operations are underpinned by various information systems that are widely in use throughout the university mainly for the management of the teaching and learning processes, students' affairs, human resources, finance, and institutional management. As a large academic institution operating within a complex business environment, UTHM requires the strategic application of information systems to maximize productivity and operational efficiency.

The primary goal of this project is to enhance the Enterprise Architecture (EA) of UTHM to achieve great alignment between business processes and information system. The project aims to improve operational effectiveness and establish a framework for improved coordination across academic, administrative, and support systems by examining the current enterprise environment and using a detailed architectural approach.

1.2 Problem Statement

UTHM has successfully migrated approximately 70% of its major academic, administrative, and support systems to a new platform. However, the new university infrastructure of various faculties and departments has not been fully integrated, creating challenges in the integration process. The use of independent systems and the partial integration of different systems can lead to critical issues, including data redundancy, having inconsistent workflows, and delays in delivery services.

In addition, UTHM's strategic ICT goals strengthening the infrastructure, integrating applications, and developing the workforce, in spite of their setting, are still not fully accessible due to the absence of a structured EA approach, it becomes challenging to make sure that all the systems are synchronized with the university's operational demands, strategic goals, and sustainability in the long run. Therefore, the further development of Enterprise Architecture (EA) is required to achieve full system, hassle-free business operations, and increased efficiency at UTHM.

1.3 Project Objectives

The main objective of this project is to analyze, design and propose an enterprise-level system architecture for UTHM by applying enterprise architecture principles and structured system design methodologies. This initiative aims to address the complexity of managing multiple academic, administrative and support systems within a university environment by providing a proper and well-aligned architectural design.

The specific objectives of this project are as follows:

1. To analyze the existing information systems at UTHM to support key university operations such as academic management, teaching and learning, administrative processes and student services.
2. To design an Enterprise Architecture for UTHM using the TOGAF framework with emphasis on the alignment between business processes, data, applications and supporting technologies.
3. To propose an integrated system design that improves data consistency, system interoperability and operational efficiency across UTHM's academic and administrative functions.
4. To provide architectural structure reference to support future system development and improvement initiatives at UTHM

1.4 Project Scope

This project focuses on the analysis and design of an enterprise-level system architecture for UTHM with emphasis on improving the alignment of business processes with information systems. The scope of this project is defined to ensure the study remains focused on enterprise architecture design while remaining feasible within the given timeframe and academic requirements.

1.4.1 In-Scope

The scope of this project includes the following:

1. **Organization analysis:** Examination UTHM's structure and core operational areas, particularly academic management, teaching and learning activities, administrative functions and student support services.
2. **Identification stakeholders:** Analysis of key users including students, academic staff, administrative staff and university management and their interaction with information systems
3. **Architecture design:** Development of enterprise architecture based on the TOGAF framework and the Architecture Development Method (ADM).
4. **System analysis:** Evaluating existing systems currently used at UTHM such as academic management systems, learning management systems, student assessment systems, staff related systems and student support systems.
5. **Gap analysis:** Comparison between the existing system environment (As-Is) and the proposed enterprise architecture (To-Be) to improve integration of the system, management of data and operational efficiency.
6. **Modelling:** Development of relevant system models and diagrams, including enterprise architecture diagrams, system architecture diagrams, use case diagrams and data models to support the proposed system design.

1.4.2 Out of Scope

The scope of this project excludes the following:

1. Full-scale system development, deployment or maintenance in a real operational environment.
2. Large-scale data migration, performance optimization, advanced system tuning.
3. Complete system coding for the whole enterprise systems beyond limited prototypes or demonstration purposes.
4. Security penetration testing, load testing and user acceptance testing.
5. Ongoing system monitoring, maintenance and post-deployment support activities.

1.5 Project Importance

The importance of projects is listed as follows:

1. **Integration:** It integrates all the fragmented university systems into a unified architecture by removing information silos.
2. **Resource optimization:** It minimizes data redundancy and work duplication in departments and optimizing daily operations
3. **Better decision making:** It improves decision making by providing accurate, real-time data which does not have any redundancy.
4. **Meet strategy:** It facilitates the digitalization objectives of the university and the long-term ICT strategy.
5. **User experience:** It enhances accessibility and usability of services to the students and staff.

Chapter 2: Literature Review

2.1 Introduction

This chapter presents a detailed literature review related to the design and development of a University Enterprise Architecture (UEA) in the context of an enterprise-level information system. It starts by defining important concepts of Enterprise Architecture (EA) including its contribution to enhance alignment of organizational processes, technology, and information to institutional goals. The chapter discusses also past research on the EA applications in the institutions of higher learning such as the University of Lampung to appreciate frequent issues such as system fragmentation, redundancy of data and absence of integrated academic or administrative platforms

The chapter then examines the chosen subsystems pertinent to the proposed university environment including student information, human resources, finance, subject registration, research management, learning management, and analytics dashboards and points out how other institutions have carried out similar subsystems. Lastly, it describes the technologies employed in EA development and some of them are database solutions, development frameworks, enterprise systems and architecture standards such as TOGAF and Zachman Framework. This chapter forms the theoretical and technical basis upon which the design and implementation of the enterprise system offered is going to be developed.

2.2 Enterprise Architecture (EA)

2.2.1 Definition and Concepts of EA

Enterprise Architecture (EA) is a comprehensive framework or blueprint that defines structure, processes, information and technology needed to effectively achieve an organization's business goals and objectives (Ardoq, 2024). In other words, the concept of EA is generally understood as a strategic field as well as a practice that allows an organization to streamline its business goals, processes, information systems, and technology infrastructure to achieve efficiency and flexibility in the long term. Rather than a distinct process per department, EA assists in crafting a high-level plan, to have all of the business elements working towards the objectives simultaneously.

The Zachman Framework and TOGAF organize EA into four major areas:

1. **Business Architecture:** Determine business governance, strategy, business processes and capabilities.
2. **Data Architecture:** Explain the logical and physical data resources, their structure and management and flow within the organization.
3. **Application Architecture:** This includes applications, interaction and alignment of applications with business processes.
4. **Technology Architecture:** This is the description of the technology infrastructure such as hardware, networks and platforms which support applications, data and business processes.

EA provides both an “as-is”, meaning the current state and “to-be”, which means the future state of the organization (Tamm et al., 2011). This serves as the basis for organizational analysis and strategic transformation. Through this blueprint, EA identifies gaps, streamline processes and prioritizes investments and initiatives that support the desired target of enterprise evolution.

In simpler terms, EA is a management approach that helps an organization in remaining structured, coordinated and dynamic. It brings together all parts of the organization from core processes, technology, people and strategies to ensure each component works smoothly with the rest. As the organization’s goals or environment change, EA serves as guidance for the evolution of the enterprise so the organization can remain competitive and effective.

2.2.2 Review of Existing Studies Related to University Enterprise Architecture

Previous research on EA in higher education emphasizes the necessity for scalable and efficient subsystems to support complex academic and administrative services in the institution. Existing literature demonstrates universities all over the world have transitioned from silos information systems to unified enterprise systems to improve operational efficiency and institutional decision making.

Case study: University of Lampung (Unila), an Indonesian public university located in Lampung (Nama et al., 2017). The university has developed and designed a more flexible and modern IT infrastructure which to address the challenges in increasing management complexity, scattered and separated systems and was driven by the university’s ambition to rank among Indonesia’s top 10 universities by 2025 (Nama et al., 2017). One of the analyses of the value chain of Unila developed core processes in various aspects in education, research, and community service, assisted by administrative processes like student admission and alumni services. Under each of these aspects there are several processes such as Students Admission, Research Activity and Alumni Services, which altogether support the institutional operation.

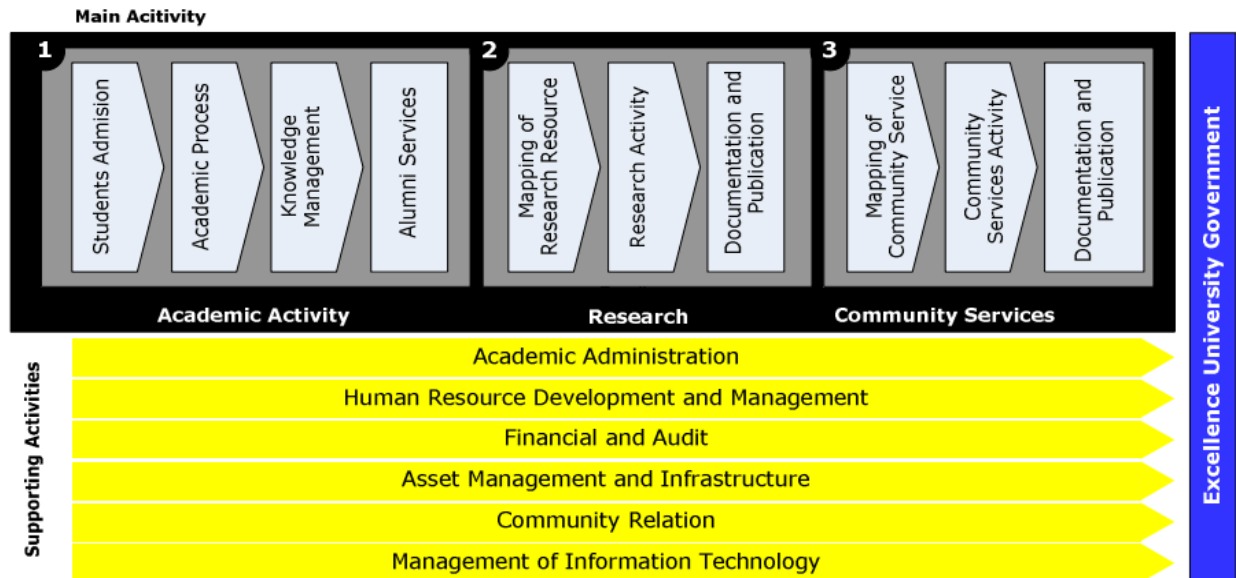


Figure 1.1: Value Chain Analysis at University of Lampung (Nama et al., 2017)

The preliminary assessment has shown that Unila had a highly fragmented IT infrastructure, with the academic management, financial and human resource system functioning independently. This absence of integration led to inefficiencies, redundancy of data and hindrance to strategic decision-making. To correct this, the research came up with a detailed architecture of integrated modular applications. Among the major improvements, the development of SIAKAD-T (an integrated academic system), a centralized library system, unified administrative platforms, Single Sign-On (SSO) solutions, and a Data Warehouse to facilitate the decision-making process (Nama et al., 2017). These solutions were developed with the aim of integration, scalability, and streamlined operations.

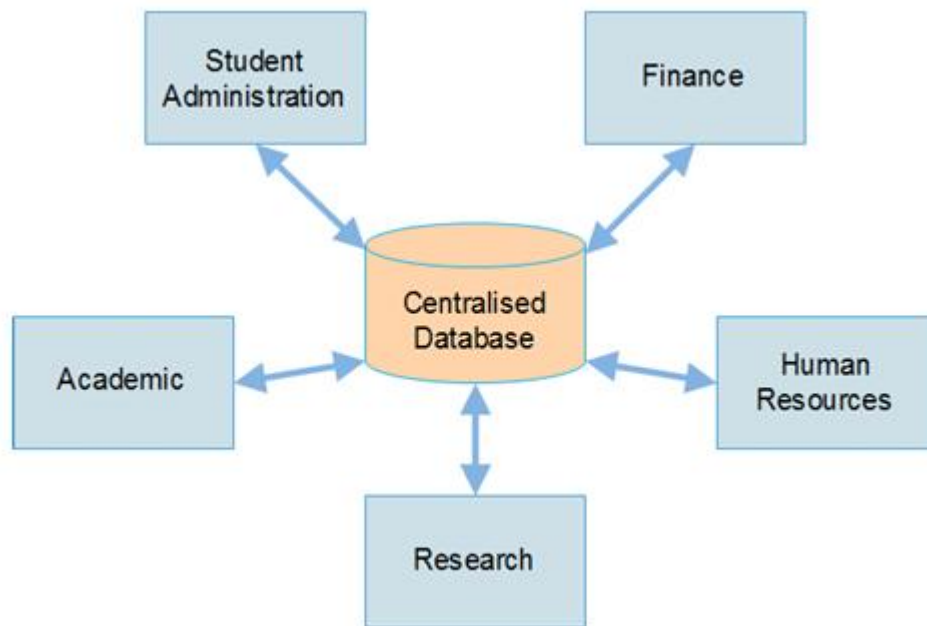


Figure 1.2: Functional Areas of a University Enterprise System at Papua New Guinea Using a Centralized Database (Daniel, 2016)

Case Study: Universities in Papua New Guinea Daniel (2016) investigated Universities in Papua New Guinea, with most of them having isolated information systems that focus on a single area of operation like admissions or finance. This segregation often led to data redundancy, inconsistent records and inefficient cross-departmental. The research suggested deployment of enterprise systems with the use of centralized databases and integrated modules of software to work with admissions, enrollments, payroll and procurements. These enhancements allow real-time information sharing and minimize data redundancy among the departmental modules (Daniel, 2016).

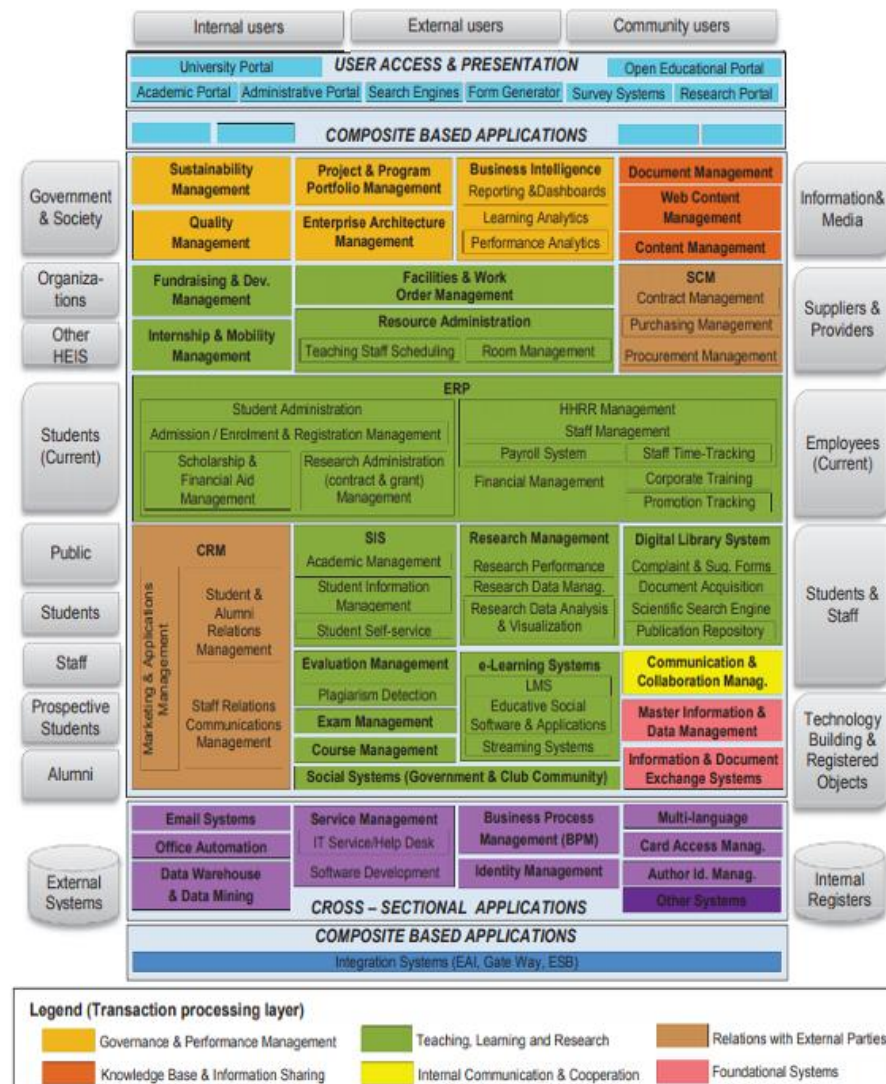


Figure 1.3: Enterprise Architecture Framework for Higher Education Institutions (Sánchez-Puchol, 2021)

The study provides a comprehensive analysis of enterprise architecture in higher education institutes (HEIs) and identifies the common challenges faced by these institutions. It is found that most universities struggle with fragmented technological environments, as these often result in inefficiencies when managing student information, academic processes, financial data, and research activities. Modern universities have adopted relevant subsystems to combat these challenges. These include student administration platforms, learning management systems, research management modules, and analytics dashboards. It is evident from the case examples provided in the study that the implementation of effective enterprise architecture has improved data accuracy, reduced administrative load, and enhanced decision support in the institution when these subsystems are properly integrated (Sánchez-Puchol, 2021).

2.3 Related Sub-Systems

This section explains the chosen sub-systems and the previous system that uses similar architecture.

2.3.1 Overview of the chosen sub-systems

The chosen subsystems for the University Enterprise Architecture are as follows:

1. Student Information System

This subsystem manages student details such as names, identification numbers, contact details, and emergency contact. It also maintains the academic records including enrollment history, course registrations, examination results, and academic achievements. The system also stores documents such as admission letters, certificates, transcripts and keeps track of graduation requirements.

2. Inventory Management System

This subsystem manages all physical assets and resources across the university. This includes necessary equipment for learning like computers, laboratory instruments, furniture and teaching aids used across all faculties. The system handles purchase requests, tracks vendor information and procurement history. It also manages asset allocation to all faculties.

3. Finance Management System

This subsystem handles all financial operations such as managing student fees based on their programs, processing tuition fees and generates receipts and invoices automatically. It also handles payroll for staff.

4. Human Resource Management System

This subsystem maintains thorough records of employees including personal details, qualifications, employment history, performance evaluations, administrative personnel and support staff. It handles recruitment workflows from job posting to hiring a candidate. The system also manages approvals for annual leave, sick leave, and study leave and payroll calculations including salaries, allowances, deductions and benefits.

5. Subject Registration System

This subsystem manages the course registration process for each semester. This includes course codes, descriptions, credit hours, and prerequisites. It makes sure no section reaches the capacity limit and allows for add or drop courses feature for students. The system also handles the approval process from the academic advisor to proceed with the subject registration.

6. Research Management System

This subsystem tracks all research activities conducted within the university including managing research project proposals, approvals workflow, research timelines and records research grants. It also tracks postgraduate student research including thesis proposals, progress reports, defense schedules and generates research performance metrics for reporting.

7. Learning Management System

This subsystem provides a digital platform for course delivery and online learning activities. This allows lecturers to upload and organize course materials including lecture notes and presentations. It also facilitates online assignment submission where students can upload their work by the deadlines and requirements specified by the lecturer.

8. Dashboard and Analytics System

This subsystem centralizes all subsystems to provide real-time insights for decision-making across the institution. It tracks key metrics such as enrollment trends, retention rates, graduation rates, research output, finances, and satisfaction. The system supports forecasting, resource optimization, and risk analysis. It also automates reports for strategic monitoring through clear visualizations like charts and heat maps.

These 8 subsystems collectively form a comprehensive university enterprise architecture that supports all major functions needed from a higher education institution. Each subsystem has its own specific functionality but must integrate seamlessly with other subsystems as well to ensure the university's operations can be run efficiently, have consistent data throughout the system and provide sufficient support and meet the university's requirements.

2.4 Technology Used

This part included major technologies that are applicable in the design of the University Enterprise Architecture (EA). SAP Business Technology Platform (SAP BTP) is going to be the primary focus in the development of the subsystem. SAP BTP is a unified Platform-as-a-Service (PaaS) which offers a single application development, data management and processes automation environment. It is selected for its ability to bridge the gap between legacy enterprise systems and modern cloud requirements.

2.4.1 Unified Data Management and Analytics

The SAP BTP uses SAP HANA Cloud as the main database technology. This memory database can enable the university to handle a lot of student and administrative information on the fly. It keeps the integrity of complex academic records intact, using sophisticated multi-model features. The subsystem can easily retrieve all the data of various sources without physical migration through the SAP BTP services such as SAP Datasphere. This effectively guarantee “Single source of Truth in the EA of the university.

2.4.2 Application Development and Integration

The development frameworks within SAP BTP facilitate both Pro-Code and Low-Code environments meanwhile the traditional frameworks like Node.js and Java are supported through the Cloud Foundry environment. This site puts emphasis on the SAP Cloud Application Programming Model (CAP) that advocates the best practices to make the subsystem maintainable and scalable. SAP BTP’s Integration Suite also provides pre-built connectors for enterprise systems to reduce the complexity of API management and allow seamless data flow between departmental applications.

2.4.3 Extension of Enterprise Systems

Core systems Core systems like the SAP S/4HANA or Microsoft Dynamics is used to manage sensitive financial and student information in a university setting. The Side-by-Side Extensibility allows developers to build new features for the university subsystem on SAP BTP without the need to modify the code of the core ERP. This is to ensure that the core system inside ERP remains clean, facilitate the future upgrading process and reduce the risk for the institution.

2.4.4 Alignment with EA Frameworks

The SAP BTP platform is designed to support structured EA frameworks like TOGAF and Zachman Framework by providing a standardized platform for four architecture layers which are business architecture, data architecture, application architecture, and technology architecture. This is to ensure the digital processes are aligned with institutional policies and have consistent governed data handling.

2.5 Summary

The literature reviewed in this chapter highlights the importance of effective and coordinated enterprise architecture to support the operational issues of contemporary universities. The findings show that institutions that have fragmented systems experience inefficiencies, inconsistencies in data, and low decision-making capabilities. EA offers a strategic roadmap in solving such problems by defining business, data, application, and technology layers. The study of other related subsystems shows that core academic and administrative operations must be aligned to ensure smooth performance of the institution. Additionally, sustainable development of the system is made possible using relational databases and enterprise solutions such as SAP BTP that has the scalability, security and interoperability it needs to support the development of the system. Consequently, this literature confirms the direction of the project design, as it proves that the implementation of EA principles supported by appropriate technologies will allow the university to progress in terms of efficiency, integration, and digitalization.

Chapter 3: Methodology

3.1 Introduction

This chapter outlines the methodological framework adopted for the development of the University Enterprise Architecture (UEA) for UTHM. It provides a guide for the approach taken throughout the project's lifecycle. This chapter explains the chosen methodology and the justification for choosing the methodology, an explanation of each phase in the methodology and a project planning schedule which governs the timeline and task distribution.

3.2 Chosen Methodology

Waterfall Methodology was chosen as the primary development methodology for this project. The Waterfall approach as a linear sequential life cycle model will have distinct phases of analysis, design, implementation, testing and maintenance which are non-overlapping (Senarath, Udes S., 2021). This follows a strict sequential approach, and each phase must be completed before starting the next as the model makes sure that the structure remains stable through the entire development lifecycle.

The approach is especially beneficial to the UTHM Enterprise Architecture project because of the systematic approach that makes it easier to handle the complicated architectural requirements. Moreover, Waterfall model requires strict documentation of all its steps where a complete documentation of system specifications and design decisions will be maintained (Samaratova A.K., 2024). The clearly defined phases will therefore enable proper estimation of the timeline and the allocation of resources to reduce chances of project overruns.

3.3 Phases of Methodology

The Waterfall methodology for the UTHM University Enterprise Architecture follows a linear and systematic approach. Each phase serves a foundational purpose to ensure that all architectural requirements are validated before moving for the next step.

3.3.1 Planning Phase

The Planning Phase is foundational basic stage where the project scope, objectives and feasibility are defined. During this phase, the project team identifies the key stakeholders at UTHM and establishes a detailed project timeline to ensure better organization and strategic planning throughout the project lifecycle. All the tasks are gathered into the GitHub Backlog enable real time tracking of the project roadmap and monitor early progress.

3.3.2 Analysis Phase

The Analysis Phase focuses on gathering and documenting the functional and non-functional requirements. This phase involves data collection through questionnaires that are distributed during industrial visits for the staff and stakeholders of UTHM. This data collection provides a clear understanding of the current operational state ("As-Is"), allowing for an accurate assessment of existing business processes and system limitations.

3.3.3 Design Phase

The Design Phase involves converting the requirements into architectural models to identify the To-Be Enterprise Architecture. This stage is associated with the design of the target system structure, identification of suitable services in the SAP Business Technology Platform (BTP), and mapping of the services to subsystems. The strategic alignment of this architecture will make it scalable and highly compatible with the various operations requirements of UTHM.

3.3.4 Implementation Phase

The Implementation Phase is where the actual establishment of the subsystem using SAP Business Technology Platform (BTP) occurs. In this phase, each team member is responsible for developing and configuring their assigned subsystem. This includes using the SAP BTP cockpit to build systems or applications, set up the databases and create integration flows specific to their function.

3.3.5 Testing Phase

Testing Phase is essential to prove the integrity and performance of the architecture. This stage involves unit testing, which is used to confirm that every SAP BTP subsystem is working properly independently and then integration testing which is used to determine that all the sub systems work together to achieve the requirements of the university.

3.4 Project Planning Schedule

Project Planning Schedule Project planning schedule is a formal plan, which organizes the different activities and the different deadlines associated with the activities in order to make the organization efficient and on schedule. Based on the Waterfall technique, a Gantt chart (see Figure 3.1) is used to visualize the project and partition the workflow into five different phases mentioned above.

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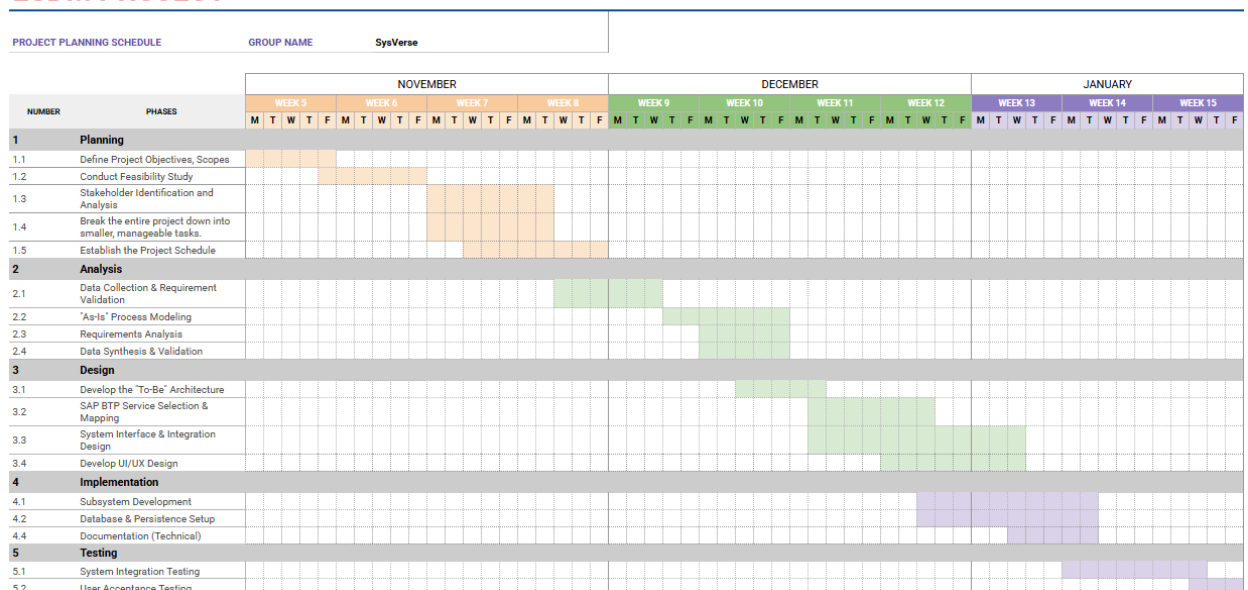


Figure 3.1 Gantt chart of ESDM Project

3.5 Summary

The project includes the use of the Waterfall Methodology, linear life cycle model, which has specific and defined stages. The approach behind this methodology is its logical simplicity, high documentation requirements and ability to deliver the correct estimates of the timeline, which are critical in the complicated development of the UTHM Enterprise Architecture. The development process is structured in five systematic stages including planning, analysis, design, implementation, and testing. Each phase is conducted by dividing the subtask into a certain time until it is successfully accepted by the stakeholder.

Chapter 4: Analysis and Design

4.1 Introduction

This chapter presents the analysis and design phase of the proposed Enterprise Architecture (EA) for Universiti Tun Hussein Onn Malaysia (UTHM). Building upon the findings from the previous chapters, this chapter focuses on examining the current organizational and system environment, identifying existing issues, and proposing a structured system design that aligns with the university's strategic digital transformation goals.

The analysis section provides an overview of UTHM's organizational structure, existing information systems, and current operational processes. This includes both computerized and manual workflows that support academic, administrative, and student support functions. By analyzing the current "As-Is" architecture, gaps and limitations related to system integration, performance, and data consistency can be identified.

Subsequently, the design section introduces the proposed "To-Be" enterprise architecture based on the TOGAF framework. This includes the design of business architecture, information systems architecture, and technology architecture. Special emphasis is given to the Hostel Management System, specifically the Room Allocation Engine subsystem, which demonstrates how enterprise architecture principles can be applied to design an integrated, rule-based, and automated solution. Various system models such as use case diagrams, activity diagrams, sequence diagrams, database design, and class diagrams are presented to clearly illustrate the proposed solution.

Overall, this chapter serves as a bridge between conceptual enterprise architecture planning and detailed system-level design, ensuring that the proposed solution is practical, scalable, and aligned with UTHM's operational and strategic requirements.

4.2 Organization Structure

The organizational structure of University Tun Hussein Onn Malaysia (UTHM) plays a critical role in shaping how information systems are planned, implemented, and governed across the institution. As a public university, UTHM operates through a hierarchical organizational structure consisting of top management, academic faculties, administrative departments, and supporting units. Each unit has specific responsibilities that collectively support the university's core functions of teaching, learning, research, and student services.

At the strategic level, the university management is responsible for setting institutional goals, policies, and digital transformation directions. These decisions influence the development and governance of enterprise systems, ensuring that information technology initiatives align with national higher education policies and the university's long-term vision. Supporting this level are central administrative departments such as academic affairs, finance, human resources, student affairs, and information and communication technology (ICT), each of which relies heavily on information systems to manage daily operations.

The ICT department plays a key role in managing UTHM's enterprise systems and infrastructure. It is responsible for system integration, data management, security governance, and the maintenance of both on-premise and cloud-based platforms. Through the implementation of centralized systems and application programming interfaces (APIs), the ICT unit ensures that data flows smoothly across different subsystems while maintaining system reliability and security.

In the context of the Hostel Management System, multiple organizational units are involved, including student affairs, hostel administration, finance, and security services. These units interact with the system to manage student accommodation, room allocation, fee collection, maintenance requests, and reporting. Understanding this organizational structure is essential for designing an enterprise architecture that supports cross-departmental collaboration, reduces redundancy, and improves service delivery.

By aligning the proposed enterprise architecture with UTHM's organizational structure, the system design ensures that roles, responsibilities, and system access are clearly defined. This alignment enables effective governance, supports operational efficiency, and facilitates future system expansion as the university continues its digital transformation journey.

4.3 Current System Analysis

The current system uses the existing technological and operational landscape of the Universiti Tun Hussein Onn Malaysia (UTHM) to move toward a fully digital system. The UTHM digital transformation strategy focused on driving the cultural change toward digital literacy, enhancing the end-to-end user experience, and updating the infrastructure with the latest technology.

4.3.1 Computerized Process and Infrastructure

The UTHM digital operations are powered by a modern, web-centric architecture supported by high-availability on-premise infrastructure. The university maintains on-premise data centers designed to Tier 3 standards, featuring dual power sources, Uninterruptible Power Supply (UPS), and generator backups. These critical facilities are located on the upper floors of the data center to protect against environmental risks such as flooding. The university also transformed from a legacy client-server architecture to a new unified system called CENTRAL, which utilizes the web-based technology. This allows lecturers, students and staff to access core modules via a standard web browser by eliminating the accessibility barriers that have been found in the previous system. A centralized master database manages the core student records while the UTHM Learning Management System (LMS), like AUTHOR platform, is responsible for storing all academic materials in a separate database container. These systems are linked through the university-developed APIs and Single Sign-On (SSO) to synchronise students' credentials and monitor in real-time. UTHM also secured their network by utilizing a dual-firewall strategy and Web Application Firewalls (WAF) to defend multi-directional attacks especially from China and Europe.

4.3.2 Manual Processes and Governance

Despite the successful move to web-technology, several workflows still remain manual and semi-automated. Certain general administrative workflows still require the physical completion of forms such as Pusat Kesihatan Universiti (PKU) in UTHM still relies on the traditional manual methods for the patient appointment system. The introduction of any new system or application is strictly regulated by a formal review process as the stakeholders are required to fill a complete Feasibility Study Requirement (FSR) document and need to be approved by the responsible committee. This is to ensure the proposed system is necessary and compatible with the university's goals before the development process.

4.3.3 Identified System Issues

During the student registration for co-curricular subjects, the system expected to handle more than 1000 concurrent users at a time. This peak load performance causes the system to be slowed-down in approximately five minutes of lag times. UTHM also states that the "We Care" support system frequently receives complaints regarding the forgotten passwords for WiFi authentication and SSO which indicate a need for more streamlined identity recovery.

4.4 Comparison between Existing and Proposed System

4.4.1 Current Digital Infrastructure

UTHM has established a comprehensive digital foundation built upon industry-standard technological components. The university operates a Tier-3 standard data center with modern web-based architecture and multi-layered security protocols, including dual firewalls and Web Application Firewalls. The centralized information system (CENTRAL) and learning management system function as integrated platforms through university-developed APIs and Single Sign-On capabilities.

This infrastructure has enhanced digital accessibility across the institution. However, several operational workflows remain dependent on manual processes, particularly administrative documentation and academic advisor appointment scheduling systems.

4.4.2 Operational Challenges

The current system exhibits notable performance limitations during high-demand periods. When concurrent user access exceeds 1,000 individuals, such as during co-curricular subject registration, the system experiences significant performance degradation, resulting in approximately five-minute response delays.

Additionally, the "We Care" support channel receives persistent password-related inquiries, indicating inadequate user identity recovery mechanisms within the existing framework.

4.4.3 Proposed System Enhancement

The proposed digital ecosystem addresses these operational deficiencies through comprehensive infrastructure improvements. The enhancement plan includes advanced load balancing mechanisms, performance optimization protocols, and potential hybrid cloud integration to accommodate increased concurrent usage.

Key development areas include:

- 1) **Complete workflow automation:** Digital transformation of remaining manual processes, including online academic advisor appointments and electronic administrative forms
- 2) **Streamlined governance processes:** Implementation of fully digital approval workflows, such as online FSR submissions and automated committee procedures
- 3) **Enhanced user experience:** Development of a unified platform interface with improved identity management capabilities
- 4) **Advanced security features:** Integration of self-service password recovery systems and multi-factor authentication options

4.4.4 Expected Outcomes

The proposed system represents a significant advancement from the current semi-digitized infrastructure. The enhanced platform will deliver improved scalability, comprehensive automation, strengthened security protocols, and optimized performance capabilities. This transformation will create a more cohesive digital environment that better serves the university community's operational and academic needs.

The transition from the existing system to the proposed infrastructure reflects a strategic move toward a fully integrated, performance-driven digital ecosystem that can effectively support the university's growing technological demands.

4.5 System Requirements Gathering Techniques

We conducted an interview session in the industrial talk to gain a deeper understanding of UTHM's daily operations, enterprise architecture, and the practical challenges faced by the ICT team and users.

The questions are categorized below:

4.5.1 Daily Operations & Current Process

1. Can you describe the main tasks you perform daily using the current system?
2. Although APIs have been adopted to integrate the data, there is still manual transfer of data between the Main Student Database and the LMS. What are the technical or procedural obstacles that would not allow total automation?

4.5.2 Enterprise Architecture & Integration

1. Now that the migration of the legacy TCIS to the web-based CENTRAL architecture has reached completion of 70-80% is already made, what do you believe are the largest technical challenges that could be expected to be encountered during the remaining 20%?
2. Given that the LMS is designed to be independent and exist as a database separate on the central system, how can data consistency and integrity be ensured so that one cannot find out inconsistency including the absence of student records?

4.5.3 Infrastructure & Performance

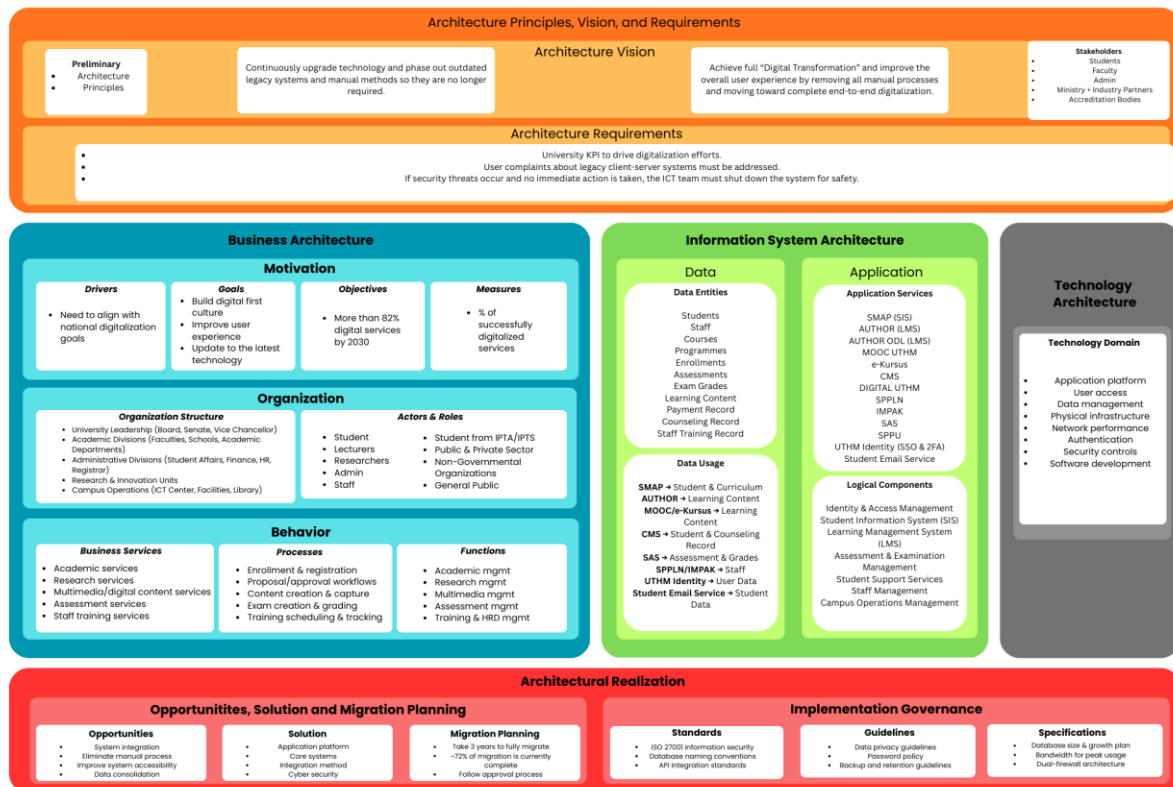
1. The latency was observed to have a time of 5 minutes during peak Course Registration (CRS) intervals that have more than 1000 simultaneous users. What is the solution to this problem by the UTHM team?
2. You explained the 3 tier power backup (TNB sources A/B, UPS, Genset). How much time will the existing UPS solution support the critical servers until the genset goes online to replace all its power?
3. Why is the Data Centre located upstairs despite the logistical convenience of a ground-floor location for loading and unloading equipment?

4.5.4 User Experience & Future Needs

1. How does the cybersecurity team protect sensitive data from cyberattack?

4.8. System Design

4.8.1 Enterprise architecture EA-AS IS



4.8.2 Explain each component in system architecture EA-AS IS

1. Architecture Principles, Vision and Requirements

a. Preliminary Principle

The main order of action is focused on gradually upgrading technology and phasing out outdated legacy systems and manual methods. This is the current transition stage of UTHM in which the institution is transitioning out of an old client-server architecture to modern web-based architecture, which is still ongoing.

b. Architecture Vision

The current vision emphasizes achieving full digital transformation by improving the overall user experience through the removal of manual processes and moving toward complete digitalization. At present, UTHM is at approximately 70-80% digital migration, meaning there is still a significant portion of operations that rely on manual or semi-automated processes.

c. Architecture Requirements

The AS-IS requirements focus on three critical areas:

1. University KPI alignment: Ensuring systems support the institution's key performance indicators and digitalization goals
2. Legacy system user complaints: Addressing issues users face with outdated client-server systems that require specific configurations
3. Security protocols: If security threats occur and no immediate action is taken, the ICT team must have protocols to shut down the system for safety

2. Business Architecture

This layer defines the existing posture of UTHM in its operations in terms of business, including organisational structure, processes, and stakeholder interactions.

A. Motivation Layer

Drivers: The existing system is driven by the need to comply with the national digitalisation goals that are set by the Ministry of Higher Education. UTHM is aware of the fact that the outdated technology and manual procedures limit the level of operational performance and undermine user experience.

Goals:

1. Create a digital-first culture within the institution.
2. Improve the experience of both faculty and students.
3. Improve infrastructure using the latest technology to ensure that the company is up to date.

Objectives: Its main agenda is to achieve over 82-percent digital services by 2030 and thus shifting paper-based and manual workflow to complete automated digital services.

Measures: Success is currently measured using the level of services that have been successful in digitalisation. UTHM keeps track of the volume of processes moved on-line and the adoption of the new web-based CENTRAL system by the users.

B. Organization Layer

The current organisational structure of UTHM is made of:

- University Management: Vice Chancellor, Board of Directors.
- Functional Divisions: Organized in terms of faculties, schools, and academic departments.
- Administrative Departments: Student Affairs, Finance, Human Resources and Registrar departments.
- Support Units: Research and Innovation Unit, ICT Centre, Facilities management, and Library.

Actors & Roles:

There are several stakeholders that the system is serving at present:

- Students: For IPTA (public universities)/IPTs (private institutions): primary users.
- Academic members of staff who present instructional content are known as lecturers.
- Researchers: Faculty taking part in scholarly research work.
- Administrative Personnel: Personnel who are involved with operational management.
- Personnel: General support staff.
- External users: partners in the public and non-governmental organisations as well as the general population to receive stated services.

C. Behaviour

The business services already available are:

- Academic services (enrolment, registration, grading).
- Research services (project management, grant tracking).
- Student services (counselling, support services).
- Evaluation services (examinations, evaluations).
- Staff trainer services (professional development).

Processes: Some of the major business processes that are currently in operations are enrollment and registration, proposal approval, content development and acquisition, exam construction and scoring, training organization and monitoring.

Functions: The fundamental functional areas include course scheduling, planning curriculum., research management, multimedia control, evaluation administration, staff development through training and HRD management.

3.Information System Architecture

A. Data Architecture

In data architecture, the core data and how these data relate to the others in information systems of UTHM are carefully established. Students, Staff, Courses, Programmes, Enrollments, Assessments, Grades, Learning Content, Payment Records, Counseling Records, and Staff Training Records are some of the most significant academic and administrative entities present in the existing AS-IS environment. These organizations in totality support the academic delivery, student services and the overall operations of the institution.

This architecture makes the utilisation of the data easier through various systems that exchange and receive data to protect the continuity of operation. Student Management and Academic Planning

(SMAP) is the key source of information on students and curriculum related information. The learning materials are delivered and shared on AUTHOR, MOOC UTHM and e-Kursus. The Content Management System (CMS) handles the student documentation and the data relating to counseling. The results of the assessment and records of the academic performance are stored in the Student Assessment System (SAS). Information related to staff is monitored by SPPLN and IMPAK and the authentication of users is centralised by UTHM Identity, therefore, facilitating Single Sign-On (SSO). Moreover, there is provision of Student Email Service that is used to facilitate official communication associated with student records.

B. Application Architecture

Application architecture outlines the essential systems that will be implemented to assist UTHM academics and administration processes. This includes primary applications of SMAP as the Student Information System (SIS), AUTHOR and AUTHOR ODL as Learning Management Systems (LMS), and additional applications, including MOOC UTHM and e-kursus to deliver long-distance learning. CMS, DIGITAL UTHM, SPPLN, IMPAK, SAS, and SPPU are all the administrative and support functions that are facilitated. UTHM Identity imposes identity management and secured access complemented with institutional email services.

Logically, the applications are grouped in functional components, which are Identity and Access Management, Student Information Management, Learning Management, Assessment and Examination Management, Student Support Services, Staff Management and Campus Operations Management. All these elements together form a unified ecosystem that supports the main academic and working processes at UTHM.

5. Technology Architecture

The technology architecture of UTHM mainly focuses on transitioning from a legacy client-server model to a modern, fully web-based platform. This is to ensure that their technology has ubiquitous access for students and staff in the university. UTHM has a high-availability infrastructure that is built on a very solid foundation. It has three internal data centers that place compliance and internal control above cloud services. In an order of priority for business continuity, UTHM has put in place a three-tier power redundancy system made up of a dual-power source, Uninterruptible Power Supplies (UPS) for immediate backup, and Gensets for long-term outages. Security, being an essential factor, UTHM has adopted a multi-brand dual-firewall architecture and Web Application Firewall (WAF) which altogether protect the institutional portals. This layer also standardizes API integration and Single Sign-On (SSO) to facilitate a seamless communication between the centralized core database and peripheral learning systems like AUTHOR.

6. Architectural Realization

This layer explains the opportunities, solution, and migration planning, and the implementation governance that UTHM has implemented.

A. Opportunities, solutions, and migration planning

Opportunities: This section identifies the key improvements that UTHM must enable to derive from their strategic goals and identified problems. The institution focuses on the system integration and eliminating the manual processes to reduce the administrative work, aiming for an End-to-End (E2E) digital goal by 2030. UTHM also prioritizes improving their system accessibility by replacing the legacy client-server architectures with fully web-based systems and ensuring their data consolidation by maintaining a single, centralized database linked to separate systems like AUTHOR.

Solutions: The solution defines the final desired state for UTHM's technology, application and data architecture. The application platform must be fully web-based for ubiquitous access, while core systems utilize a centralized system for the core data and AUTHOR as the LMS. The integration method of UTHM relies on the API and SSO supported by the UTHM data center infrastructure which focuses on the internal infrastructure and compliance. For reliability, UTHM maintains a 3-tier power redundancy and a multi-brand security layer featuring dual firewalls alongside a Web Application Firewall.

Migration planning: This outlines the strategy to execute the transition to the target state. UTHM has utilized a "face-to-face" migration strategy where the old system runs concurrently with the new implementation. The UTHM has estimated their migration duration is about 3 years as they have reached approximately 72% of completion. All new applications must strictly follow the UTHM approval process including filling out a complete MSR requirements and committee check.

B. Implementation governance

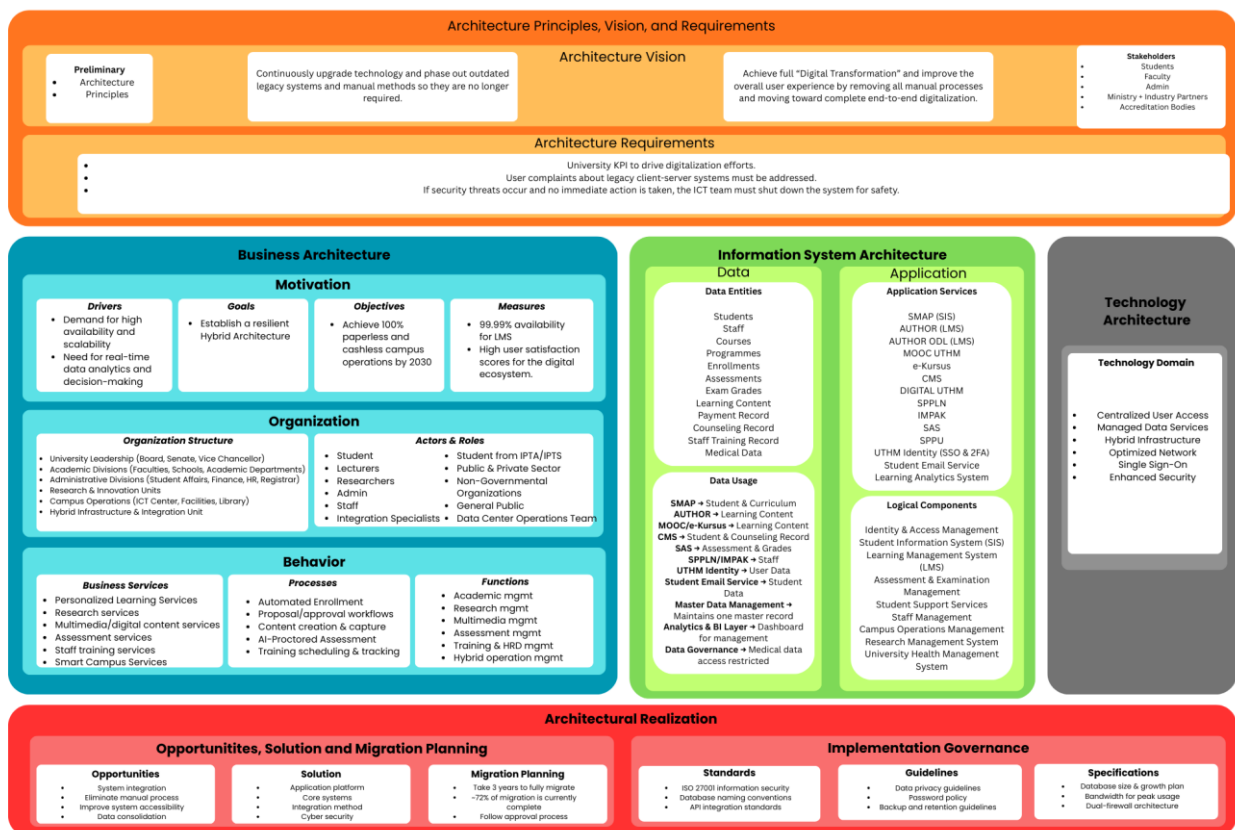
Standards: This component is to define a formal, mandated criteria to ensure the quality and consistency across UTHM systems. The ISO 27001 information security helps the UTHM to formalize the commitment to robust their security systems. UTHM also mandates the database naming conventions to manage the centralized but segmented data environment and API integration standards to ensure the smooth data transfer between the core database and the LMS.

Guidelines: Data privacy guidelines helps the UTHM to ensure compliance with PDSA standards for sensitive student data and academic marks. A structured password policy is established to

address user experience issues meanwhile the backup and retention guidelines are implemented in UTHM to protect the centralized data and ensure infrastructure continuity.

Specifications: These include the database size and growth plan to ensure the centralized system does not degrade as the university moves toward 2030 digital goals. The UTHM has also addressed their bandwidth for peak usage specifically targeting bottlenecks that have been observed during the student co-curricular registration.

4.8.3 Enterprise architecture EA-TO BE



4.8.4 Explain each component in system architecture EA- TO BE

The elements of a TO-BE architecture of a university are usually based on the TOGAF layers but are directed to optimization and future possibilities. There are some of the components in the EA TO-BE:

1. Architecture Principle, Vision, and Requirements

This layer determines the objectives of the transformation at the high level.

i. Architecture Principle:

Architecture Principles are the rules and guidelines, which guide UTHM in developing and managing its IT resources. The principles are focused on interoperability, which will mean that different systems such as Finance and Academic records can communicate with each other without issues, and agility, which will enable the infrastructure to expand as the university grows. These principles are focused on a user-centric concept and they make sure that all the technological developments are meant to enhance the performance and experience of students and staff instead of complicating their day to day activities.

ii. Architecture Vision:

The Architecture Vision serves as the North Star to the university and is the description of the way it wants to be fully integrated into Digital UTHM. The first objective is to become 82% digital by 2030, to move to the modern digital ecosystem, frontrunner of fragmented legacy systems and paper-based workflows. The vision will focus on ensuring that there is a single experience where all stakeholders can use a single secure portal to access all the university services and create a culture of data-drivenness that will ensure the mission of excellence of TVET education is achieved by the university.

iii. Architecture Requirements:

Architecture Requirements are measurable, specific, "must-haves" which the system has to fulfill in order to be declared successful. They comprise a high level of compliance with security measures, including the use of Multi-Factor Authentication (MFA) and ISO 27001 standards to keep safe institutional data. Also, these requirements will guarantee that the IT strategy is optimally aligned with the Ministry of Higher Education (MOHE) KPIs and directly responds to user pain points, as the new applications should be stable, high-performing and address the real needs of the campus community.

2. Business architecture

This layer explains the strategy, people, and operations needed for UTHM's future Hybrid Architecture.

A. Motivation Layer

Drivers: The university faces external pressure to provide systems that are always online and can grow with student demand. Additionally, the management team needs real-time data analytics to make instant decisions instead of relying on slow, manual reports. These drivers force UTHM to move beyond basic websites to powerful, scalable platforms.

Goals: The primary goal is to establish a Hybrid Architecture that combines the security of on-premise servers. This ensures that UTHM data is safe while maintaining the speed and accessibility of academic applications.

Objectives: UTHM is also striving to become 100 percent paperless and cashless by the year 2030. This will imply the removal of all the physical forms of activities and cash transactions in order to streamline daily activities. This will save a lot of administrative time and efficiency.

Measures: The measure of success will consist of reaching the 99.99 percent system availability, which means that the Learning Management System (LMS) is hardly ever down. High user satisfaction rates are another important indicator, and it demonstrates that the new digital ecosystem is not a futile project to students and employees. These measures mean that the technology will provide actual value and not novel features.

B. Organization Layer

In order to facilitate the new hybrid architecture, we have introduced a new Hybrid Infrastructure and Integration Unit to take care of the key relationship between the on-premise servers at UTHM and the cloud. This department will work alongside current departments but will add new professional skills such as Integration Specialists whose duty will be to make sure that the system is not only interoperable but also with a Data Center Operations Team that focuses on the physical Tier 3 infrastructure.

C. Behaviour

Business Services: We incorporate Personalized Learning Services whereby the content is adjusted to the needs of individual students rather than having a general lesson. Besides that, we include a new service known as Smart Campus Services which uses IoT technology to access such facilities as smart parking and smart libraries. This changes the emphasis on the delivery of basic services to intelligent and user-friendly experience.

Processes: We replaced the manual processes with Automated Enrollment whereby we could regulate instantly without the delay of the human approval. The introduction of AI-Proctored Assessment is also aimed to provide the control of online examinations, which are autopiloted and avoid bodily invigilators. These procedures are more focused on expediency and automation as opposed to manual data input.

Functions: Hybrid Operation Management has now been incorporated into the management functions in order to manage the cloud services and the on-premise hardware at the same time. This provides a single perspective of the overall IT health of the university. It enables IT teams to easily and proactively manage problems in various environments.

3. Information System Architecture

The Information system architecture level connects between the business strategy at the high level and the technology. It consists of two major fields: Data Architecture that describes what needs to be known, and Application Architecture that describes the computer programs that manage that information.

a) Data architecture

Data architecture involves defining important data objects and the flow of information within the organization in order to maintain data integrity and security.

The most basic units of the UTHM information ecosystem are data entities. It could be subdivided into their particular role:

- Essential Academic Branches: Students, personnel, courses, and programs.
- Academic history: classifications, examinations, and grades.
- Operational Assets: Learning material and payment history.
- Sensitive Records: Special training and medical history records.

The movement of data between systems is called data usage. It involves the data being utilised to bring value to the various stakeholders.

- LMS Integration: Learning Management Systems draw on student and curriculum information to load the online classrooms.
- Administrative Management: Content Management Systems (CMS) is used to align student records with learning materials.
- Security & Identity: Identity systems are used to control credentials of users to make sure the correct individuals can access the correct data.
- Strategic Intelligence: Business intelligence (BI) tools combine data to provide a management dashboard.
- Governance: The medical data access is highly divided and limited to the privacy policies.

b) Application architecture

Application architecture is a concept that is used to describe the software services and logical modules, which facilitate user interaction with the data.

Application services is the functional software platforms deployed:

- Learning Platforms: SMAP, AUTHOR and MOOC LMS suites.
- Administrative Platforms: CMS and e-Kursus in managing the curriculum.
- Enterprise Services: Learning Analytics, Identity management and Email systems.

Logical component represents the central driver of the digital operations of the university:

- Identity & Access Management: Authentication and user rights.
- Student Information System (SIS): This is the database of student lifecycle.
- Learning Management (LMS): The teaching, learning, and collaboration interface.
- Assessment and examination: Intensity course modules: grading and exam integrity.
- Campus Operations: Systems: Research and day-to-day logistical systems.
- Health Management: A safe university medical service system.

In general the comparison between the information system architecture of EA as-is and EA to-be is that the To-Be architecture added completely new modules that the As-Is does not have, namely the University Health Management System. Then, implements a special Master Data Management layer and Data Governance framework to manage limited access, especially of medical records. Finally, includes an Analytics and BI Layer to convert the raw data into actionable dashboards to the university management.

4. Technology Architecture

This layer defines the software and hardware infrastructure required to support the deployment of business, data, and application services. The TO-BE technology architecture for UTHM shifts from fragmented, legacy hardware to a resilient, high-performance hybrid environment.

i. Hybrid Infrastructure

The core of the TO-BE model is a Hybrid Cloud environment. This combines UTHM's on-premise Tier 3 Data Center for sensitive localized data (like medical and financial records) with Scalable Cloud Services such as AWS or Azure for student-facing applications like the LMS and MOOC. This ensures the university can handle massive traffic spikes during enrollment or exam periods without system crashes.

ii. Centralized Architecture & Managed Data Services

Unlike the previous decentralized "silo" approach, the TO-BE architecture utilizes Centralized Management. All server resources, storage, and databases are governed under a unified framework. Managed Data Services are implemented to automate backups, database tuning, and high-availability clustering, ensuring that the "Master Data" remains consistent across all campus departments.

iii. Optimized Network & Connectivity

To support the "Smart Campus" and IoT goals (smart parking/libraries), the network infrastructure is upgraded to High-Density Wi-Fi 6 and a 10Gbps Fiber Backbone. This optimization ensures low latency for AI-proctored assessments and seamless connectivity for thousands of simultaneous mobile users across the UTHM campus.

iv. Single Sign-On (SSO) & Identity Management

To solve login fatigue issue faced by students and staffs, the technology layer implements a Unified Identity Provider (IdP). Using protocols like SAML or OpenID Connect, the Single Sign-On (SSO) allows students and staff to access the LMS, Finance, HR, and Medical systems using a single set of UTHM credentials, significantly improving the user experience and reducing password-reset tickets.

v. Enhanced Security (Zero Trust & Dual-Firewall)

Security is integrated directly into the infrastructure through a Zero Trust Architecture.

- Dual-Firewall Design: Traffic is inspected through both external perimeter firewalls and internal micro-segmentation firewalls to protect sensitive data zones.
- Encryption at Rest & in Transit: All data moving between the cloud and on-premise servers is encrypted using TLS 1.3, and databases are encrypted to meet ISO 27001 standards.

5. Architectural Realization

i. Opportunities & Solutions:

Opportunities and Solutions are the next phase where UTHM establishes the What by assessing the disconnect between the existing legacy systems and the envisioned new digital future. This element is aimed at bundling together similar business requirements into rational work packages i.e. consolidating the manual leave and claims processes into one "HR Digital Module." Within the framework of this project, it explains the manner in which UTHM decides on the outdated client-server applications that need to be removed or upgraded to achieve its 82% digitalization goal in the university.

ii. Migration Planning:

Migration Planning determines the "When" through creating a time-based roadmap of IT transformation of the university. As shown in the EA diagram, UTHM is at stage of polishing with 72 percent of the migration already accomplished as part of the 3-year strategic plan. The component is aimed at having digital projects prioritized effectively depending on their available budget as well as the urgency of security threats and accomplishment of critical university KPIs, so that there is a smooth transition to the target architecture.

iii. Implementation Governance:

The Implementation Governance is the How by implementing the rules and standards that make sure that new systems are safe and well-integrated. UTHM is employing ISO 27001 as a reference to the information security benchmark and fully demands technical control by means of

specifications, such as the so-called Dual-Firewall design. This element plays a policeman role, making sure that new development is done to the standards of the institution, such as requiring the use of UTHM Identity (SSO/2FA), so as to not create an insecure or fragmented system of logging in.

4.8.3 System Architecture

The Cloud System Architecture relies on the SAP Business Technology Platform (BTP). The design uses a Low-Code/No-Code strategy to ensure efficiency and scalability. The system comprises four key layers. The Presentation Layer utilizes SAP Build Apps to provide role-specific interfaces for students, wardens, and security staff. The Application Layer leverages SAP Build Process Automation to handle core logic like room allocation. Additionally, the SAP Integration Suite connects securely to external systems like payment gateways. The Data Layer centralizes all information using SAP HANA Cloud to store live records. Finally, the Analytics Layer uses SAP Analytics Cloud to visualize key metrics through dashboards. This setup ensures a complete workflow from user input to strategic reporting.

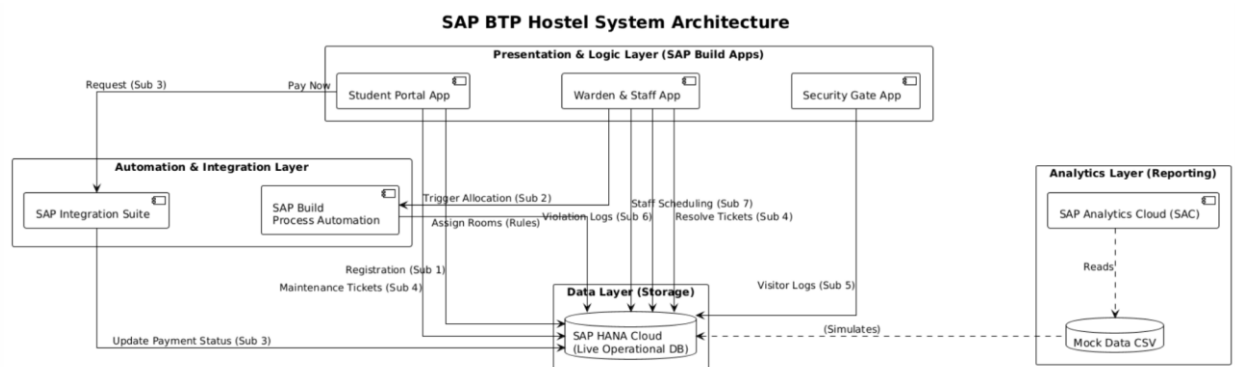


Figure 4.8.3.1 system architecture of hostel system

4.8.4 System Architecture Component Explanation

The Hostel Management System utilizes the SAP Business Technology Platform (BTP). It employs a cloud-native, low-code strategy. This approach minimizes the need for custom coding. Consequently, the system is easier to build, test, and maintain.

The architecture consists of four distinct layers:

1. Presentation & Logic Layer (User Interface)

Component: SAP Build Apps

This component functions as the unified user interface. It also handles the backend logic for daily tasks. We used SAP Build Apps to create role-based interfaces. These interfaces run smoothly on both web and mobile devices.

- **Student Portal App:** This is a mobile-friendly interface for students.
 - **Registration (Subsystem 1):** This feature captures student details. It uploads profile photos directly to the database.
 - **Payment UI (Subsystem 3):** Students view outstanding invoices here. They can also initiate payment requests.
 - **Maintenance (Subsystem 4):** Students submit tickets through this portal. The app's internal logic saves these tickets immediately. They appear with an "Open" status in the database. This process requires no separate workflow engine.
- **Warden & Staff Operations App:** This is a tablet-optimized app for management.
 - **Ticket Resolution (Subsystem 4):** Wardens view "Open" tickets. The Warden taps "Complete" when a job is finished. The app logic updates the status to "Resolved" instantly.
 - **Staff Scheduling (Subsystem 7):** Wardens view duty rosters here. They can also request shift swaps within the interface.
 - **Disciplinary Log (Subsystem 6):** Staff record student violations. These records link directly to the specific Student ID.
- **Security Gate App:**
 - **Visitor Logging (Subsystem 5):** This is a simplified scanner interface. It reads visitor QR codes. The system then logs entry and exit timestamps.

2. Automation & Integration Layer (Business Rules)

Component: SAP Build Process Automation

The apps handle simple logic. However, this component manages complex decisions requiring algorithms.

- Room Allocation Engine (Subsystem 2): This service executes the Business Rules. A student submits an application. The engine then checks Gender, Year of Study, and Room Preference. It compares this data against available inventory. Finally, the system assigns the correct Block and Room automatically. No human intervention is necessary.

Component: SAP Integration Suite

- Payment Connectivity (Subsystem 3): This suite acts as a secure bridge. It connects the hostel system to external services. A student clicks "Pay" in the app. The Integration Suite then manages the secure API handshake. It connects with Payment Gateways like FPX or JomPay. The system processes the fee and updates the record.

3. Data Layer (Central Storage)

Component: SAP HANA Cloud

This component is the centralized, in-memory database. It serves as the "Single Source of Truth."

- Unified Storage: The database stores data from all subsystems. This includes Student Profiles, Maintenance Tickets, Staff Rosters, and Visitor Logs.
- Real-Time Sync: Updates are instant because the database is cloud-based. A student may register via the mobile app. The Warden's tablet updates immediately to reflect this change.

4. Analytics Layer (Reporting)

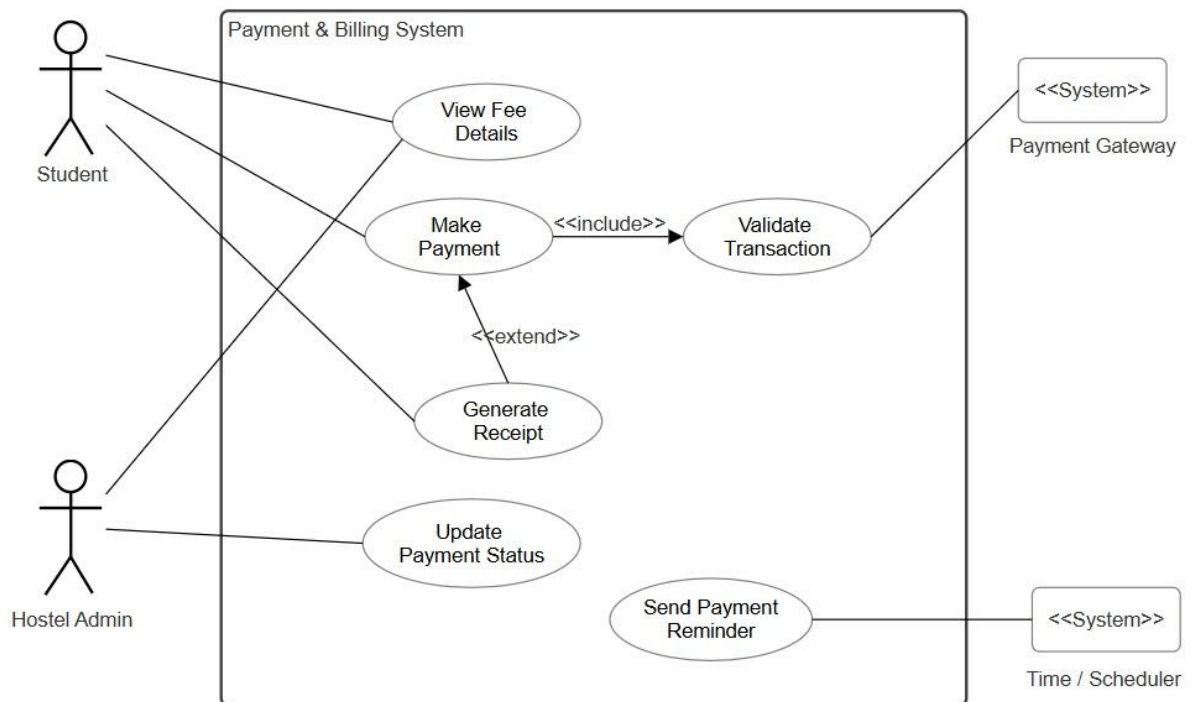
Component: SAP Analytics Cloud (SAC)

This layer provides strategic oversight for the administration.

- Dashboard & Reporting (Subsystem 8): This component connects to the data layer. It visualizes performance metrics.
 - Occupancy Trends: The dashboard visualizes hostel capacity by block.
 - Financial Health: It displays "Total Revenue Collected" versus "Outstanding Debt."
- Mock Data Implementation: The current phase uses a Mock Data Model. This allows the system to simulate high-volume historical scenarios. For example, it can display yearly trends not yet available in the live database.

4.8.5 Project Design

4.8.5.1 Use Case Diagram - Payment & Billing Tracker



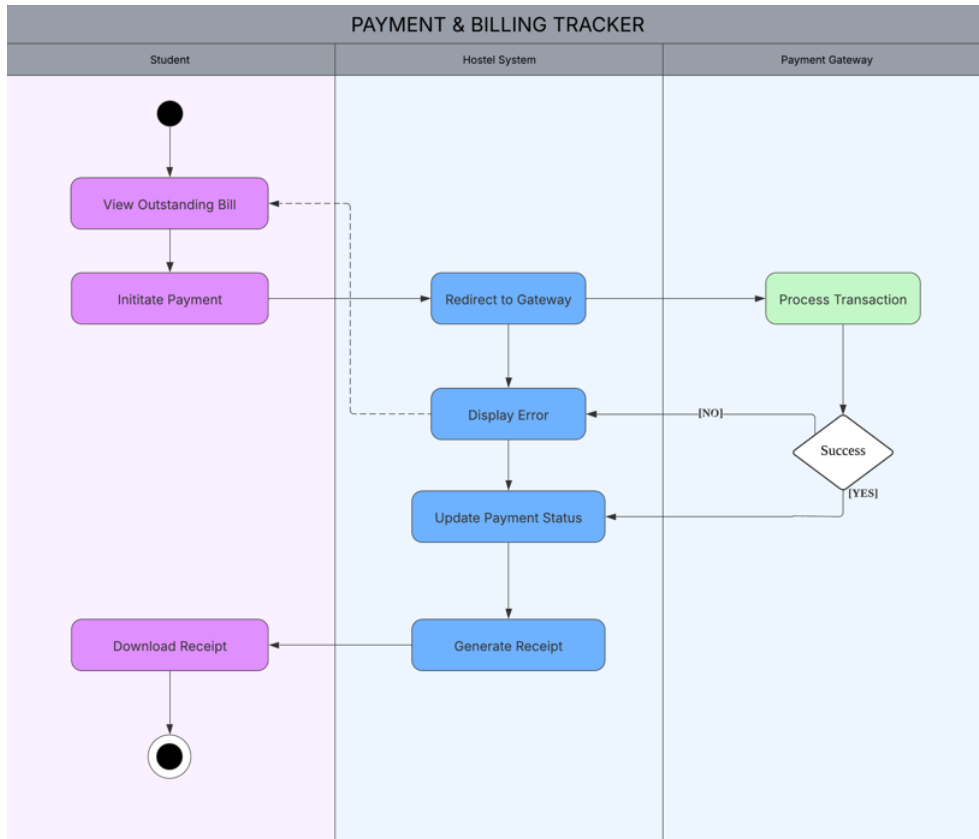
4.8.5.2 Use Case Description - Payment & Billing Tracker

Use Case Name		Payment & Billing Tracker
Actor		Student, Hostel Admin
Preconditions		1. The student must be logged into the Hostel Pay system. 2. An outstanding bill must exist for the student.
Normal Flow	Description	1. Students navigate to the "Hostel Bills" section. 2. The system displays the current outstanding balance and bill details. 3. Students initiate the payment process for the specific bill. 4. System validates the request and redirects the connection to the external Payment Gateway.

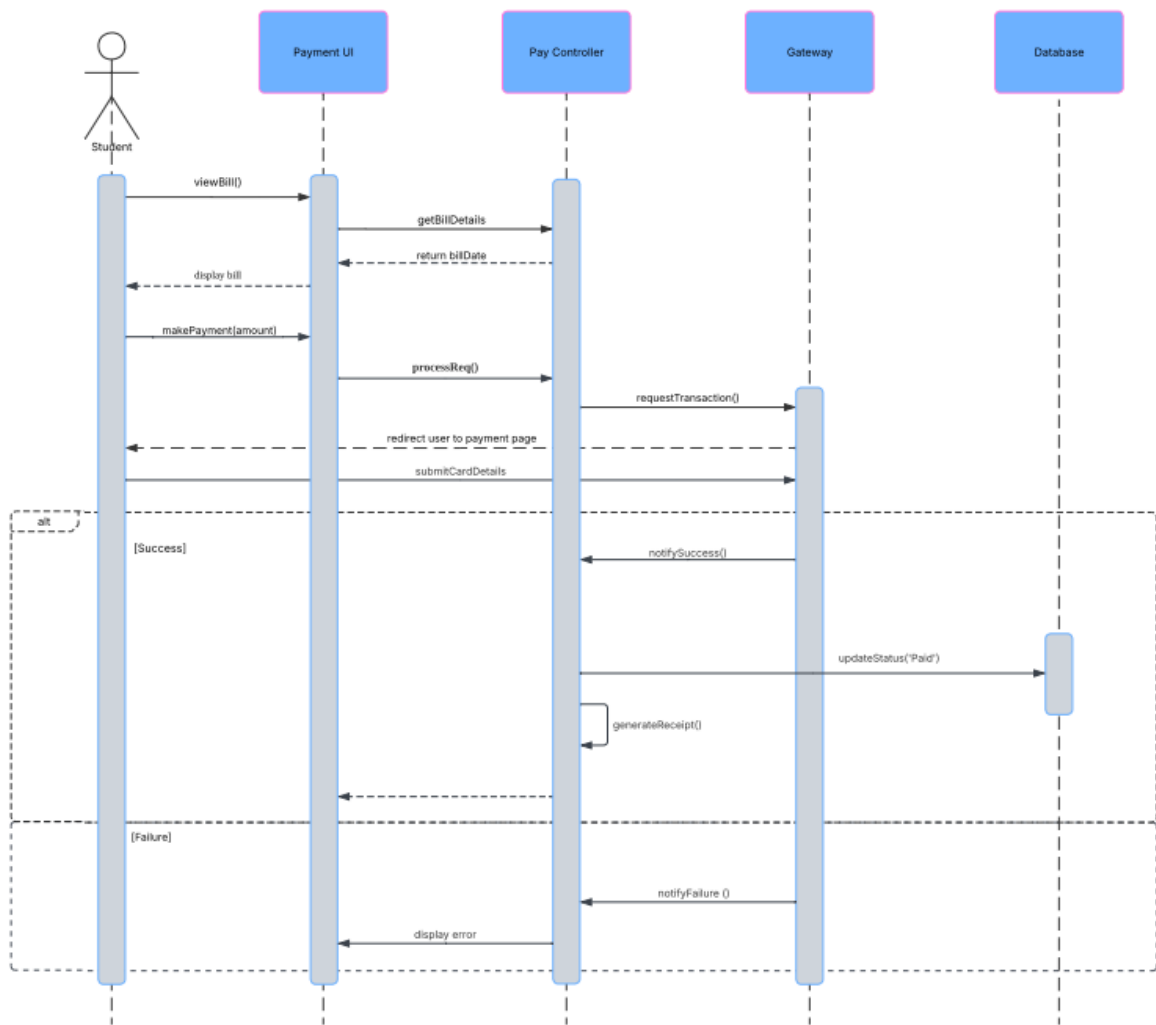
		5. Payment Gateway processes the transaction securely. 6. Payment Gateway confirms the successful transaction to the System. 7. The system records the payment and updates the bill status to "Paid". 8. The system generates a receipt for the transaction. 9. The system notifies the Student that the payment was successful.
	Postconditions	Success: The Student's outstanding balance is updated to 0, the Billing status is set to 'Paid', and a new Receipt record is created. Failure: The database remains unchanged, and the bill remains pending.
Alternative flows		A1. Payment Failed / Rejected - At Step 4, if the Payment Gateway encounters an error (insufficient funds, wrong OTP, cancelled). - The Gateway sends a notifyFailure() signal to the System. - The System displays a "Transaction Failed" error message on the UI. - The System redirects the Student back to the Dashboard. - The Bill status remains "Overdue" or "Unpaid". A2. Gateway Timeout - If the Student stays on the Gateway page for too long without action. - The session expires, and the Student is returned to the app with a "Session Timeout" notification.
Non-functional requirements		1. Security: Data encryption using TLS 1.3 for secure transmission and compliance with PDPA standards to protect student financial information. 2. Performance: The system should load the payment page within 2 seconds and support up to 1000 concurrent users during peak periods.

3. **Usability:** The interface must be fully responsive for mobile devices and provide immediate, clear feedback messages to the user.

4.8.5.3 Activity Diagram - Payment & Billing Tracker

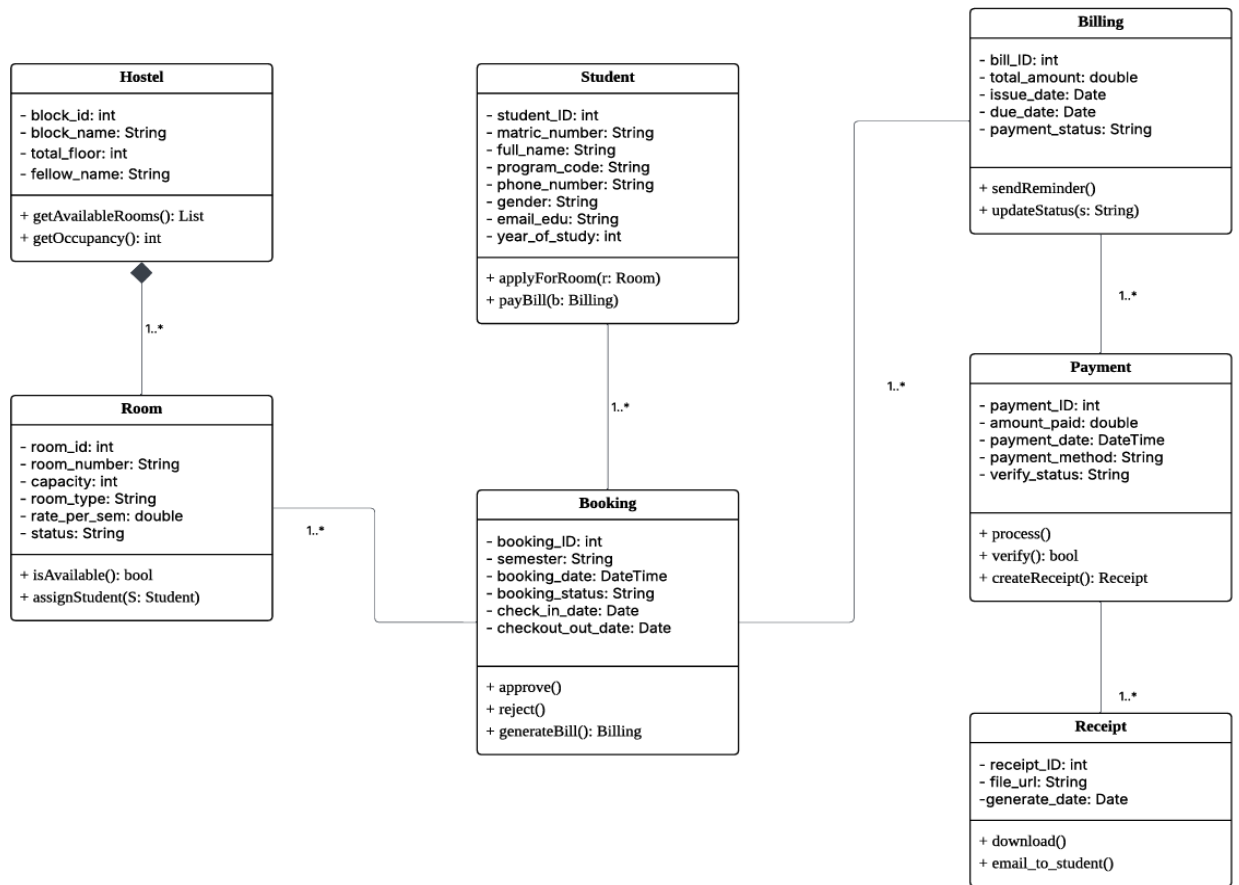


4.8.5.4 Sequence Diagram - Payment & Billing Tracker

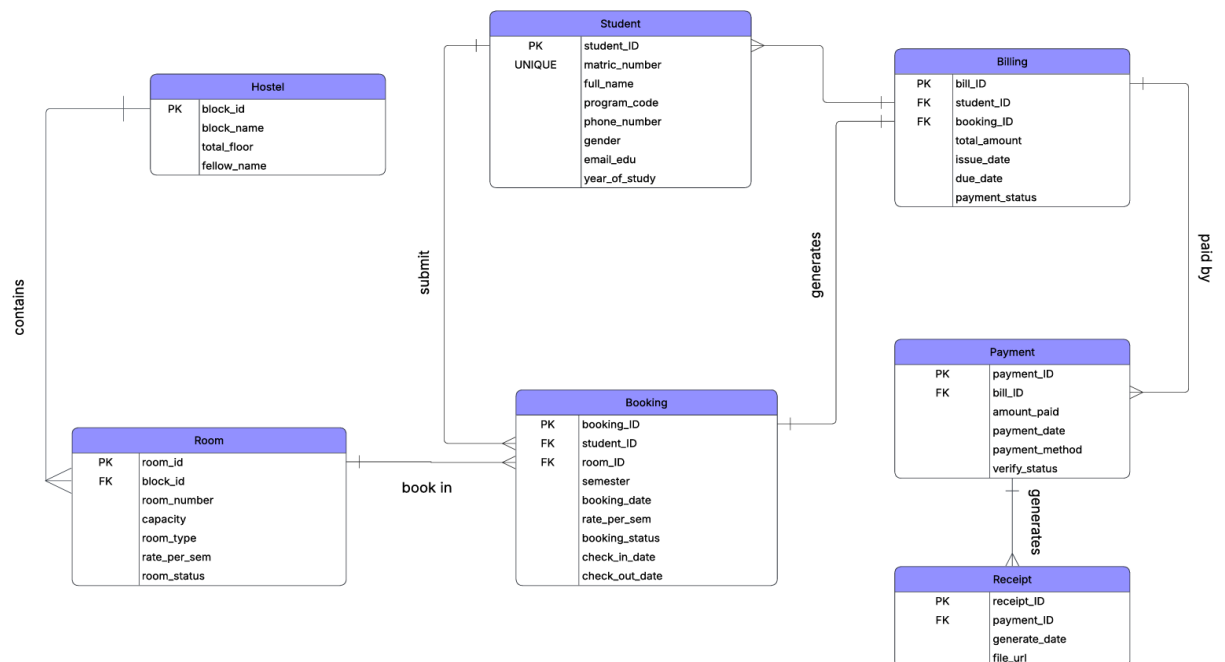


4.8.6 Database Design - Payment & Billing Tracker

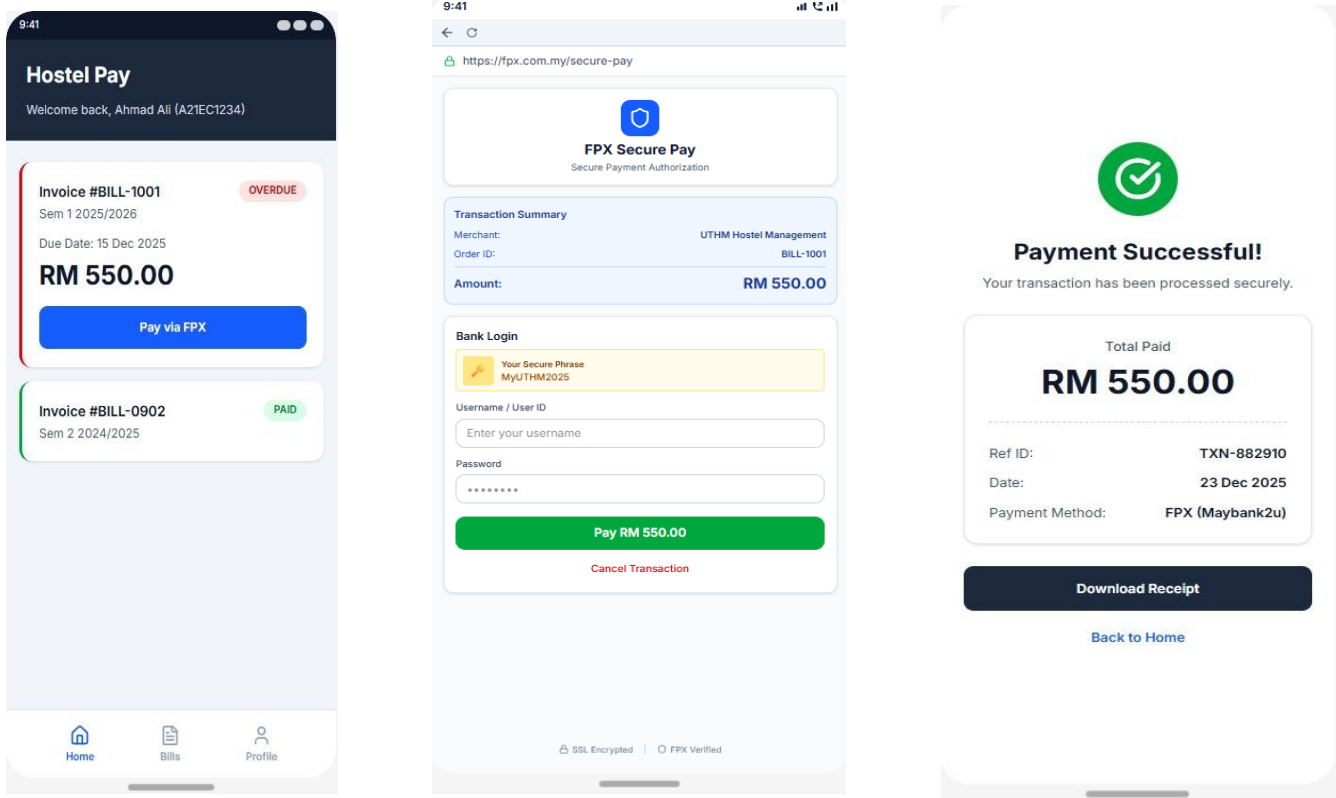
4.8.6.1 Class Diagram - Payment & Billing Tracker



4.8.6.2 Entity Relationship diagram - Payment & Billing Tracker



4.8.7 Interface Design - Payment & Billing Tracker



4.9 Summary

This chapter has provided and analysed a detailed analysis and design plan of the UTHM Enterprise Architecture project. It commenced with the evaluation of the shortcomings of the system and operational issues of the existing system, and then the functional and non-functional requirements were defined based on the UTHM industry talk. The chapter proceeded to describe the strategic change between the As-Is and the To-Be architecture, which is an SAP BTP cloud-based solution. Lastly, the Payment & Billing Tracker subsystem was described in detail with the use of the UML diagrams and database models to guarantee a scalable, secure, and user-centric digital solution.

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